

06 — Mission Crane 3000 kW (budget devoured, plant reshuffles)

Proves · Priority devouring the whole budget — and the anti-double-counting rule between the battery and Level 3.

Mechanics this scenario turns on

- A **Mission** row takes **H** = the heavy consumer's full rating — a crane can start at any moment, so its whole draw is a potential peak.
- Cascade per row: **H** = what it wants · **I** = $\min(\text{remaining budget}, H)$ · **J** = $I \times \text{CoverageFactor}$ (covered) · **L** = $H - J$ (left to the gensets). Priority: DpReserve → DpDemand → Mission → Propulsion → Hotel. Invariant $\sum J + \sum L = \sum H$ — the sea sets the swing, the battery moves the split.
- Level 3 (DRC) damps the hotel/mission swing: $\text{variation} \times 0.8$, minus whatever the battery already shaved, so the same kilowatt is never counted twice.

Panels described below · Battery Contribution · Power Demands — the reshuffle · Baseline & IL1 · Battery Benefit & IL3

Anything not described here behaves exactly as in [01-excel-baseline](#) — same plant, same hours; only what this scenario changes is worked through below.

Trust · verified against the reference workbook. These figures are proof.

Read after · scenario 05.

Test 01's vessel + **Mission Max = 3000** (battery 1260, Transit).

Battery Contribution

Row	H	I	J	L
Mission (D=0.5)	3 000	$\min(1260, 3000) = 1\ 260$ (ALL of it)	630	2 370
Propulsion	573.15	$\min(0, ...) = 0$	0	573.15
Hotel	76	0	0	76

Tiles: **PS = 630** · **SR = 3 019.2**. The crane, third in priority, drains the budget; everyone below starves.

Power Demands — the reshuffle

Hotel' = $3\ 800 + 2\ 370 + 76 = 6\ 246$ — MORE than one SG (3 250) can carry. Options: 1 ME+SG+AEs (SG 3 250 + AE 2 996 — expensive aux running), or **2 ME + both SGs (6 500) + 0 AE**.

The optimizer picks the second: ME = $12\ 036.15 + 6\ 246 = 18\ 282$ @ 38.1 %, **AE = 0**, SG at 96.1 % ($6\ 246/6\ 500$). Energy ME $18\ 282.15 \times 5\ 000 = 91\ 410\ 750$.

Baseline & IL1

Sorted: **3.1095 (2/on/0 — now FIRST!)** / 3.1292 (1/1) / **3.1780 (1/2 — baseline)** / 3.2102 / 3.2210. The combo that was the WORST row in test 01 is now the best — the uncovered reserve changed which configuration wins, not just how much fuel it burns.

- Baseline = $3.1780 \times 5\,000 = \mathbf{15\,890\ t/yr}$ (AE 2 996/8 000 = 37.5 %)
- IL1 = $3.1095 \times 5\,000 = \mathbf{15\,547.3}$ · **Savings 342.7 t = \$267 335** — 16× test 01's, because the candidate rows are now genuinely different machines, not ± 1 idle AE.

Battery Benefit & IL3

Benefit = covered 630 monetized: **631.3 t/yr = \$492 407** — the ceiling for this battery (630 = the whole 1260 × 50 %). IL3 detail: the mission covered band (630, hotel side) fully absorbs the ± 500 DRC variation → Variation shows **0**, batteryShaved **500**, L3 adds **0** (anti-double-counting, rule Q4/D4).

Takeaway: the cascade runs BEFORE combination enumeration for a reason — uncovered reserve can flip the optimal plant configuration entirely.

KSailCalc manual test scenarios — calculation walkthrough · regenerated 2026-08-07 · numbers verified against commit 559aef2 and unchanged by the backend/client refactors (18 golden snapshots byte-identical)